

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No-12	Team No-17	Project Title-Inventory & Warehouse Management System
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1. Abstract

Modern supply chains face severe challenges in managing inventory efficiently due to issues like stockouts, overstocking, and fulfillment delays caused by manual tracking methods. This project proposes an integrated Inventory & Warehouse Management System designed to centralize and automate multi-warehouse inventory operations in real time

The system leverages Barcode Scanner API integration for swift, error-free stock inwarding, order picking, and dispatching. Core capabilities include automated reorder thresholds to trigger supplier purchase orders, multi-warehouse stock transfer logging, and comprehensive supplier lifecycle management. Additionally, an interactive analytics dashboard provides operational visibility into inventory valuation, turnover rates, and demand forecasts. By replacing fragmented, manual workflows with automated digital operations, the system optimizes holding costs, improves accuracy, and streamlines order fulfillment processes.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090196	M. Laasya Sri	M. Laasya Sri
2520090197	M. Rajini	M. Rajini
2520090313	K. Ramya	K. Ramya

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frontend: React.js, Chart.js**Backend:** Node.js, Express.js**Database:** MySQL**APIs / Tools:** Barcode Scanner API, JWT (Authentication)**Guide Name and Signature (remarks if any)** _____